

EDUCATION

Degree/Certificate	Institute/Board	CGPA/Percentage	Year
B.Tech. CSE	Karunya institute of Technology and Science	7.71	2023-27
Senior Secondary	Tamil Nadu State Board	86%	2022-2023
Secondary	Tamil Nadu State Board	100%	2020-2021

PROJECTS

- CAREERARC

AI Resume Intelligence Platform

GitHub

– Developed an **AI-powered resume analysis platform** for **ATS scoring, job matching, and NLP-based content optimization.**

– Built a **full-stack web application** using **React, Node.js, Express, and MongoDB** with REST API architecture.

– Integrated **LLM APIs (Gemini, Llama 3.1)** for resume parsing, rewriting, and intelligent career recommendations.

– Implemented **secure authentication, document management, and PDF report generation** for scalable user workflows.
- HydroWatch

DevOps-Enabled AI-Driven Water Monitoring Platform

GitHub

– Developed an **AI-driven real-time monitoring system** for water pipeline analysis and anomaly detection.

– Built backend services using **FastAPI** and designed a **web dashboard** for visualization and control.

– Applied **Machine Learning models (Random Forest, Isolation Forest)** achieving **94% accuracy** in leak detection.

– Integrated **Prometheus and Grafana** for monitoring and alerting via **Telegram, Email, and Webhooks**.

– Deployed using **Docker, Kubernetes, and GitHub Actions** implementing scalable **CI/CD pipelines**.
- Smart-Tag

ML-Based Toll Verification and Fraud Detection System

GitHub

– Developed a **Computer Vision-based toll verification system** using **Machine Learning techniques**.

– Implemented **vehicle detection (YOLO)** and **license plate recognition (PaddleOCR)** for real-time data extraction.

– Designed an end-to-end pipeline for **image processing, OCR, database verification, and fraud detection**.

– Built a **Flask-based web dashboard** using HTML, CSS, and JavaScript for monitoring and visualization.

– Improved toll efficiency through **automated real-time verification and anomaly detection**.
- Lost & Found Hub

Full-Stack Web Application

GitHub

– Developed a **full-stack web application** for item tracking using **HTML, CSS, JavaScript, Node.js, Express, and MySQL**.

– Implemented **secure user authentication, REST APIs, and advanced search/filtering** for efficient data management.

– Integrated **EmailJS** for automated email notifications and real-time updates on matched items.

– Designed a **responsive user dashboard** improving usability and accessibility across devices.
- Smart Study Reminder

IoT-Based Embedded System

GitHub

– Designed an **IoT-based embedded system** using **NodeMCU (ESP8266), tilt sensor, buzzer, and I2C LCD**.

– Implemented **real-time posture detection** and automated study-break cycles with audio-visual alerts.

– Utilized **non-blocking programming (millis())** for efficient multitasking and continuous system operation.

– Improved student productivity through **structured scheduling and real-time monitoring**.

TECHNICAL SKILLS

- **Programming:** Python, Java

• **Web Development:** HTML, CSS, JavaScript, React, Node.js, Express

• **Databases:** MySQL, MongoDB

• **Machine Learning:** Random Forest, NLP, LLM APIs (Gemini, Llama)

• **Computer Vision:** YOLO, PaddleOCR, Image Processing

• **DevOps & Tools:** Docker, Kubernetes, GitHub Actions (CI/CD), Prometheus, Grafana

• **Core Concepts:** Data Structures & Algorithms, DBMS, OOPs, Operating Systems